A. Bender Problem

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Robot Bender decided to make Fray a birthday present. He drove n nails and numbered them from 1 to n in some order. Bender decided to make a picture using metal rods. The picture is a closed polyline, which vertices should be nails (in the given order). The segments of the polyline should be parallel to the coordinate axes. Polyline is allowed to have self-intersections. Bender can take a rod and fold it exactly once in any place to form an angle of 90 degrees. Then he can attach the place of the fold to some unoccupied nail and attach two ends of this rod to adjacent nails. A nail is considered unoccupied if there is no rod attached to it (neither by it's end nor the by the fold place). No rod could be used twice. It is not required to use all the rods.

Help Bender to solve this difficult task.

Input

The first line contains two positive integers n and m ($4 \le n \le 500$, $2 \le m \le 500$, n is even) — the amount of nails and the amount of rods. i-th of the following n lines contains a pair of integers, denoting the coordinates of the i-th nail. Nails should be connected in the same order as they are given in the input. The last line contains m integers — the lenghts of the rods. All coordinates do not exceed 10^4 by absolute value. Lengths of the rods are between 1 and $200\,000$. No rod can be used twice. It is guaranteed that all segments of the given polyline are parallel to coordinate axes. No three consecutive nails lie on the same line.

Output

If it is impossible to solve Bender's problem, output `NO`. Otherwise, output `YES` in the first line, and in the second line output n numbers — i-th of them should be the number of rod, which fold place is attached to the i-th nail, or `-1`, if there is no such rod.

If there are multiple solutions, print any of them.

Examples

input Copy

```
4 2
0 0
0 2
2 2
2 0
4 4
```

output Copy

```
YES
1 -1 2 -1
```

input Copy

```
6 3
0 0
1 0
1 1
2 1
2 2
0 2
3 2 3
```

output Copy

```
YES
1 -1 2 -1 3 -1
```

input Copy

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 28 (Codeforces format)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
372671059	Apr/27/2026 07:54	Accepted
372670950	Apr/27/2026 07:52	Runtime error on test 1
372670789	Apr/27/2026 07:48	Accepted

```
6 3
0 0
1 0
1 1
2 1
2 2
0 2
2 2 3
```

output Copy

```
NO
```

372670733	Apr/27/2026 07:47	**Accepted**
372670650	Apr/27/2026 07:46	Compilation error
372670447	Apr/27/2026 07:42	Wrong answer on test 1

→ Problem tags

implementation *1600

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1 (en)
- Tutorial #2
- Tutorial #3 (ru)
- Tutorial #4
- Tutorial #5

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